ESP32 Air Quality Monitoring Station – Technical Documentation

1. Project Overview

This project is a compact air quality monitoring station built using an ESP32 microcontroller, a PMS5003 particulate matter sensor, and a BME280 environmental sensor.

The system measures PM1, PM2.5, PM10, temperature, humidity, and atmospheric pressure, processes the data using several filtering algorithms, and uploads the results to Google Sheets over WiFi.

Main objectives:

- Real-time indoor air quality monitoring
- Long-term stability through sensor duty-cycling
- Data logging to cloud services
- Clean and maintainable firmware architecture

2. Hardware Components

Microcontroller

ESP32 DevKit (WiFi enabled)

Sensors

Plantower PMS5003 – laser particle sensor (PM1 / PM2.5 / PM10)

BME280 – temperature, humidity and pressure sensor (I2C)

Passive Components

220µF electrolytic capacitor – power stabilization

100nF ceramic capacitor – high-frequency noise filtering

Other Components

Breadboard

JST-GH to pin adapter for PMS5003

USB power supply (5V)

3. Electrical Connections

ESP32 ↔ PMS5003

VIN (5V) → PMS VCC

GND → PMS GND

GPIO16 → PMS TXD

GPIO17 → PMS RXD

GPIO4 → PMS SET

ESP32 ↔ BME280

3.3V → BME VCC

GND → BME GND

GPIO21 → SDA

GPIO22 → SCL

Power Filtering

220µF capacitor between VCC and GND

100nF ceramic capacitor between VCC and GND

These capacitors stabilize voltage during PMS fan startup and filter high frequency electrical noise.

4. Sensor Operation

PMS5003 measures particle concentration using laser light scattering.

Workflow:

1. Air is drawn into the sensor by an internal fan

2. Particles pass through a laser beam

3. Scattered light is detected by a photodiode

4. Internal firmware calculates particle mass concentration

Measured parameters:

PM1 – particles smaller than 1 μm

PM2.5 – particles smaller than 2.5 μm

PM10 – particles smaller than 10 μm

5. Power Management – Duty Cycle

Continuous operation can cause sensor contamination and increased temperature.

Therefore the PMS5003 operates in duty-cycle mode.

Measurement cycle:

Wake sensor

30 seconds stabilization

10 seconds measurements

Sensor sleep

This reduces fan wear and prevents overheating.

6. Cleaning Cycle

To reduce particle buildup inside the optical chamber the firmware performs a cleaning cycle.

Every ~5 hours:

Sensor runs continuously for 3 minutes

This increases airflow and removes some accumulated particles from the chamber.

7. Data Processing Algorithms

Median Filter (5 samples)

Removes single outlier measurements produced by the sensor.

EMA Filter (Exponential Moving Average)

Smooths temperature, humidity and pressure data.

Humidity Correction

High humidity causes particles to absorb water and appear larger.

Correction formula:

$$PM_corrected = PM / (1 + k * RH)$$

Where:

RH – relative humidity (0–1)

k – empirical constant (~0.4)

8. Physical Validation of Particle Data

Valid particle size distribution must follow:

$PM1 \leq PM2.5 \leq PM10$

If sensor reports impossible values, firmware corrects them automatically to maintain physical consistency.

9. Data Logging Architecture

ESP32 connects to WiFi and sends measurements to Google Sheets using HTTP requests.

Data pipeline:

ESP32 → WiFi → Google Apps Script → Google Sheets

Logged columns:

Device Time

Logged Time

Temperature

Humidity

Pressure

PM1

PM2.5

PM10

RSSI (WiFi signal strength)

10. System Advantages

• Low-cost IoT air quality monitoring

• Reliable long-term operation

- Cloud data logging
- Sensor error filtering
- Modular architecture
- Expandable firmware

11. Possible Future Improvements

- Local SD card backup
- OTA firmware updates
- Home Assistant / MQTT integration
- Outdoor enclosure with airflow control
- Grafana dashboards